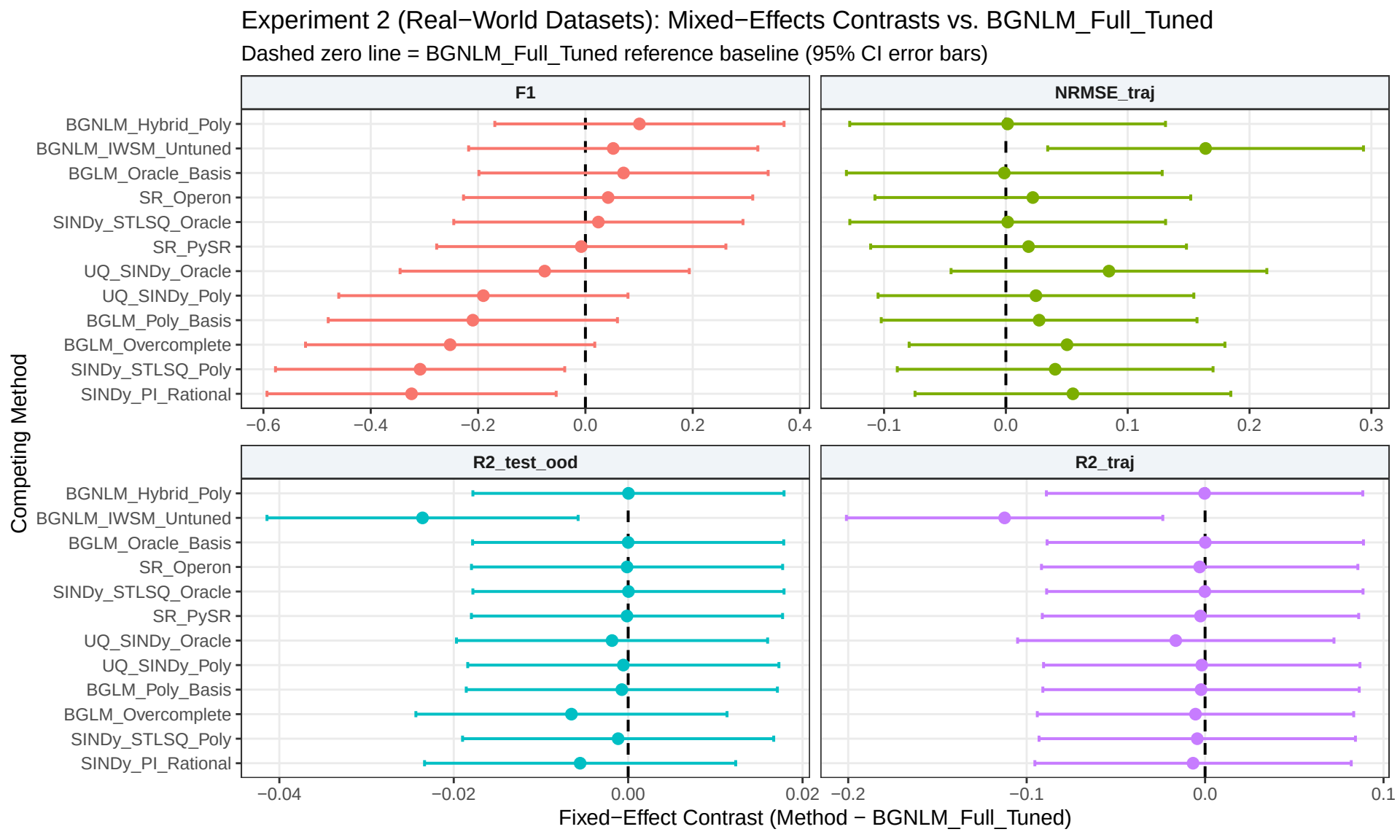
Experiment 2 (Real−World Datasets): Mixed−Effects Contrasts vs. BGNLM_Full_Tuned
Dashed zero line = BGNLM_Full_Tuned reference baseline (95% CI error bars)
F1
NRMSE_traj
R2_test_ood
R2_traj
BGNLM_Hybrid_Poly
BGNLM_IWSM_Untuned
BGLM_Oracle_Basis
SR_Operon
SINDy_STLSQ_Oracle
SR_PySR
UQ_SINDy_Oracle
UQ_SINDy_Poly
BGLM_Poly_Basis
BGLM_Overcomplete
SINDy_STLSQ_Poly
SINDy_PI_Rational
Competing Method
−0.6
−0.4
−0.2
0.0
0.2
0.4
−0.1
0.0
0.1
0.2
0.3
−0.04
−0.02
0.00
0.02
−0.2
−0.1
0.0
0.1
Fixed−Effect Contrast (Method − BGNLM_Full_Tuned)